

Final Project Report: Global Debt Analysis (2000-2024)

1. Summary

This report presents an end-to-end data analytics project analyzing the international debt of developing nations over a 24-year period (2000-2024). Utilizing data from the World Bank, the project successfully transitioned raw, unstructured CSV data into a fully normalized MySQL relational database. The data was subsequently analyzed and visualized using a custom Python/Plotly web dashboard and a comprehensive Power BI dashboard. The findings reveal a heavily centralized debt distribution and exponential growth patterns that require targeted global financial policy interventions.

2. Problem Statement

Developing nations face varying degrees of economic strain due to external debt burdens. Understanding the distribution, historical trends, and specific indicators (such as principal repayments vs. interest) of this debt is critical for global financial institutions (like the IMF and World Bank) to make data-driven decisions regarding debt relief, refinancing, and economic assistance. This project aims to provide an interactive, query-driven BI solution to answer these exact analytical questions.

3. Methodology & Tech Stack

3.1 Data Engineering (Python & MySQL)

- **Data Ingestion & Cleaning:** Raw data was ingested using Python (`pandas`). "Fake" aggregate regions (e.g., "World", "Low & middle income") were systematically filtered out to ensure accurate country-level granularity.
- **Database Normalization:** A relational database schema was designed in MySQL. The data was split into three distinct tables (`Countries`, `Indicators`, and `Debt_Data`) to achieve 3rd Normal Form (3NF), ensuring data integrity and query efficiency across over 1.5 million rows.

3.2 Analytics & Visualization

- **Python Plotly Web App:** Utilizing [SQLAlchemy](#), Python queried the MySQL database directly to generate an interactive, standalone HTML dashboard. This application features global choropleth maps, time-series line charts, and indicator distributions.
- **Power BI Dashboard:** A visually immersive dashboard was developed utilizing advanced DAX measures to dynamically format vast numerical differences (scaling automatically from Millions to Trillions) to ensure readability and professional presentation.

4. Key Analytical Insights

Through the execution of 30 specialized SQL analytical queries, several critical trends emerged:

1. **Exponential Trend Accumulation:** Global debt among developing nations remained relatively stable in the early 2000s but experienced vertical acceleration following the 2008 global financial crisis and the 2020 pandemic.
2. **Centralization of Debt Volume:** Absolute debt is heavily skewed. Large developing economies such as China, Brazil, India, and Mexico hold multi-trillion dollar debt portfolios, heavily skewing the global average.
3. **Indicator Breakdown:** "External debt stocks" make up the largest raw volume. However, "Principal repayments" and "Interest payments" demand a massive portion of national GDPs, highlighting liquidity risks.
4. **Vulnerable Extremes:** While massive nations hold the most *absolute* debt, smaller island nations and least developed countries consistently show the lowest raw debt, yet often suffer from the highest debt-to-GDP vulnerability.

5. Data-Driven Policy Recommendations

Based on the SQL querying and dashboard analysis, the following actions are recommended for global financial stakeholders:

- **Targeted Restructuring:** Because debt is centralized among the top 10 heaviest debtors, IMF and World Bank restructuring efforts focused on these specific nations will provide the most significant stabilization to the global economy.
- **Principal vs. Interest Relief:** Analysis of indicator distributions shows that refinancing programs must heavily target principal relief, as principal repayments vastly overshadow short-term interest volumes.
- **Enhanced Monitoring:** Strict monitoring and tighter lending policies are required for high-risk years and aggressive borrowing nations to prevent future sovereign defaults.

6. Conclusion

By leveraging a modern data stack (MySQL, Python, Power BI), this project successfully transformed raw economic statistics into an interactive, actionable business intelligence tool. The resulting dashboards not only answer core analytical questions but also provide an expandable framework for future global financial analysis.